

Function

$$f(x) = x^2 + 2$$

single, well defined result.

$$f(3) = 3^2 + 2 = 11$$

$$f(-1) = (-1)^2 + 2 = 3$$

In mathematics

it is a rule that assigns exactly one output to each input.

In Java

function/method it is a block of code that performs a specific task.

Types of memory

1. Stack → managing fn calls

2. Heap

Return types

```
public class Main {
    public static void main(String[] args) {
        System.out.println(args.length);
        System.out.println("Hello Akarsh!");
    }
}
```

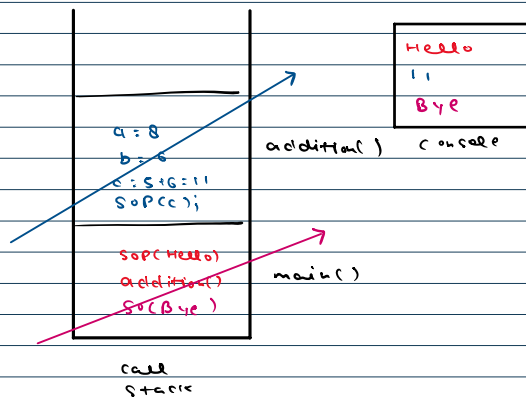
method
 ↗ Parameterized
 ↘ Non-parameterized

call stack

represent the sequence of fn calls.

- ① currently which fn is execution.
- ② how it has reached here.

```
1. public class Main {
2.     public static void main(String[] args) {
3.         System.out.println("Hello Akarsh!");
4.         addition();
5.         System.out.println("Bye Akarsh!");
6.     }
7.
8.     public static void addition() {
9.         int a = 5;
10.        int b = 6;
11.        int c = a + b;
12.        System.out.println(c);
13.    }
14. }
```



```
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
    }
}
```



```

    addition();
    System.out.println("Bye-Akarsh!");
}

```

```

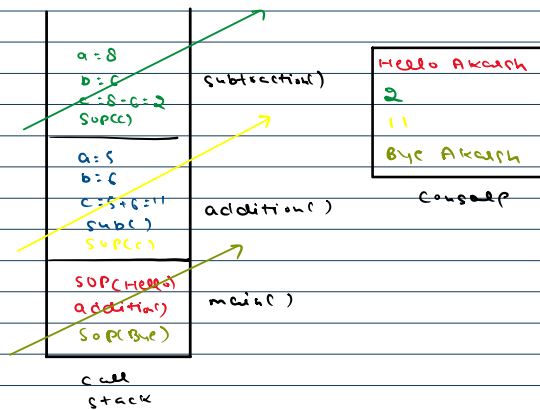
public static void addition() {
    int a = 5;
    int b = 6;
    int c = a + b;
    System.out.println(c);
}
}

```

```

1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("Hello Akarsh!");
4         addition();
5         System.out.println("Bye Akarsh!");
6     }
7
8     public static void addition() {
9         int a = 5;
10        int b = 6;
11        int c = a + b;
12        subtraction();
13        System.out.println(c);
14    }
15
16    public static void subtraction() {
17        int a = 8;
18        int b = 6;
19        int c = a - b;
20        System.out.println(c);
21    }
22 }

```



```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        addition();
        System.out.println("Bye Akarsh!");
    }
}

```

```

public static void addition() {
    int a = 5;
    int b = 6;
    int c = a + b;
    subtraction();
    System.out.println(c);
}

```

```

public static void subtraction() {
    int a = 8;
    int b = 6;
    int c = a - b;
    System.out.println(c);
}
}

```

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        subtraction(b, a);
        System.out.println("Bye Akarsh!");
    }
}

```

```

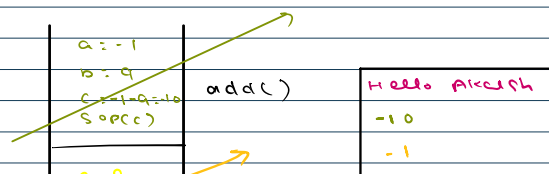
public static void subtraction(int a, int b) {
    int c = a - b;
    System.out.println(c);
}
}

```

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        subtraction(b, a);
        System.out.println("Bye Akarsh!");
    }
}

```





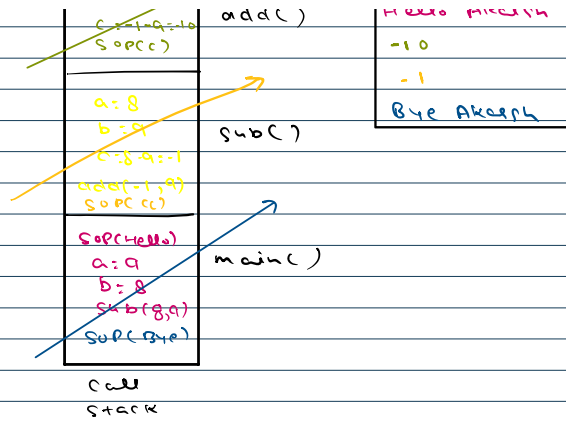
```

int a = 9;
int b = 8;
subtraction(b, a);
System.out.println("Bye Akarsh!");
}

public static void subtraction(int a, int b) {
    int c = a - b;
    addition(c, b);
    System.out.println(c);
}

public static void addition(int a, int b) {
    int c = a + b;
    System.out.println(c);
}

```



```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        subtraction(b, a);
        System.out.println("Bye Akarsh!");
    }

    public static void subtraction(int a, int b) {
        int c = a - b;
        addition(c, b);
        System.out.println(c);
    }

    public static void addition(int a, int b) {
        int c = a + b;
        System.out.println(c);
    }
}

```

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        int x = addition(a, b);
        System.out.println("Bye Akarsh!");
    }

    public static int addition(int a, int b) {
        int c = a + b;
        return c;
    }
}

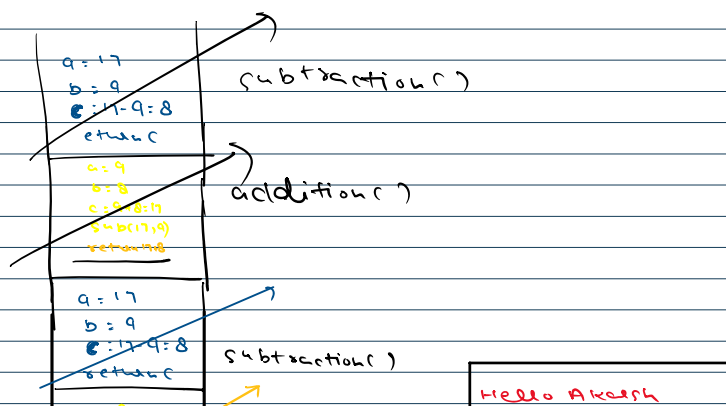
```

```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        int x = addition(11, b);
        System.out.println("Bye Akarsh!");
    }

    public static int addition(int a, int b) {
        int c = a + b;
        return c;
    }
}

```



```

1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("Hello Akarsh!");
4         int a = 9;
5         int b = 8;

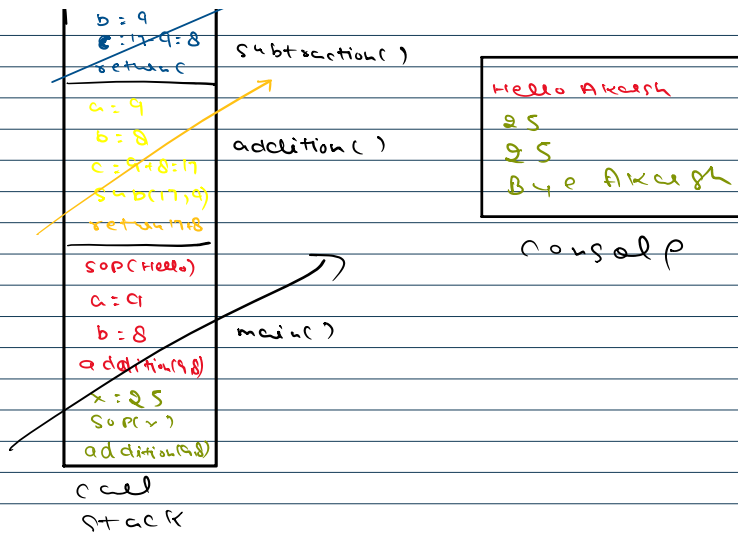
```



```

1 public class Main {
2     public static void main(String[] args) {
3         System.out.println("Hello Akarsh!");
4         int a = 9;
5         int b = 8;
6         int x = addition(a, b);
7         System.out.println(x);
8         System.out.println(addition(a, b));
9         System.out.println("Bye Akarsh!");
10    }
11
12    public static int addition(int a, int b){
13        int c = a + b;
14        return c + subtraction(c, a);
15    }
16
17    public static int subtraction(int a, int b){
18        int c = a - b;
19        return c;
20    }
21
22 }

```



```

public class Main {
    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        int x = addition(a, b);
        System.out.println(x);
        System.out.println(addition(a, b));
        System.out.println("Bye Akarsh!");
    }

    public static int addition(int a, int b){
        int c = a + b;
        return c + subtraction(c, a);
    }

    public static int subtraction(int a, int b){
        int c = a - b;
        return c;
    }
}

```

```

public class Main {

    static int val = 100;

    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        // int x = addition(a, b);
        // System.out.println(x);
        System.out.println(val);
        System.out.println(addition(a, b));
        System.out.println(val);
        System.out.println("Bye Akarsh!");
    }

    public static int addition(int a, int b){
        int c = a + b;
        val = val - 5;
        return c + subtraction(c, a);
    }

    public static int subtraction(int a, int b){
        int c = a - b;
        return c;
    }
}

```




```

public class Main {

    static int val = 100;

    public static void main(String[] args) {
        System.out.println("Hello Akarsh!");
        int a = 9;
        int b = 8;
        // int x = addition(a, b);
        // System.out.println(x);
        System.out.println(val);
        System.out.println(addition(a, b));
        System.out.println(val);
        System.out.println("Bye Akarsh!");
    }

    public static int addition(int a, int b){
        int c = a + b;
        int val = 90;
        Main.val = Main.val - 5;
        return c + subtraction(c, a);
    }

    public static int subtraction(int a, int b){
        int c = a - b;
        return c;
    }

}

```

Inverse of a number

Assume that for a number of n digits, the value of each digit is from 1 to n and is unique. E.g. 32145 is a valid input number.

Write a function that returns its inverse, where inverse is defined as follows

Inverse of 32145 is 12543. In 32145, "5" is at 1st place, therefore in 12543, "1" is at 5th place; in 32145, "4" is at 2nd place, therefore in 12543, "2" is at 4th place.

Print the value returned.

Input format

Integer

Output format

Integer

Example 1

Input
32145

Output
12543

Explanation

Assume that for a number of n digits, the value of each digit is from 1 to n and is unique.
 .g. 32145 is a valid input number.
 Inverse of 32145 is 12543. In 32145, "5" is at 1st place, therefore in 12543, "1" is at 5th place; in 32145, "4" is at 2nd place, therefore in 12543, "2" is at 4th place.

rev: 3 $\begin{matrix} \rightarrow 1 \\ \rightarrow 2 \end{matrix}$ 1 2 3 ✓
~~4 5 3~~
~~6 5 7~~
~~++2~~

3	2	1	4	5	1	2	5	4	3
5	4	3	2	1	5	4	3	2	1

3	2	1	4	5	1	2	5	4	3
5	4	3	2	1	5	4	3	2	1

4	3	7	1	5	6	2	5	2	3	7	6	1	4
7	6	5	4	3	2	1	7	6	5	4	3	2	1

3	2	1	4	5
5	4	3	2	1

↓

$$3 \times 10^4 + 2 \times 10^3 + 1 \times 10^2 + 4 \times 10^1 + 5 \times 10^0$$

$$5 \times 10^{3-1} + 4 \times 10^{2-1} + 3 \times 10^{1-1} + 2 \times 10^{4-1} + 1 \times 10^{5-1}$$

1	2	5	4	3
5	4	3	2	1

↓

$$1 \times 10^4 + 2 \times 10^3 + 5 \times 10^2 + 4 \times 10^1 + 3 \times 10^0$$

↓

$$1 \times 10^{5-1} + 2 \times 10^{4-1} + 5 \times 10^{3-1} + 4 \times 10^{2-1} + 3 \times 10^{1-1}$$

→ for (i = 0 to n)

ans pos $\times 10^{\text{digit} - 1}$

3	2	1	4	5	→	5	pos	1
---	---	---	---	---	---	---	-----	---

$1 \times 10^{5-1}$
 ...

10^{-1}

	<u>digit</u>	<u>pos</u>	<u>ans</u> $pos \times 10^{-}$
3 2 1 4 5	→ 5	1	$1 \times 10^5 - 1$
	→ 4	2	$2 \times 10^4 - 1$
	→ 1	3	$3 \times 10^3 - 1$
	→ 2	4	$2 \times 10^2 - 1$
	→ 3	5	$5 \times 10^3 - 1$

```

import java.util.*;
public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        System.out.println(inverse(n));
    }

    public static int inverse(int n){
        int sum = 0;
        int pos = 0;

        while(n > 0){
            int digit = n % 10;
            sum = sum * pos + digit * (pos-1);
            n = n / 10;
            pos++;
        }

        return sum;
    }
}

```

